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Contra-Rotating Hover-Propeller Design

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Nowadays we are witness to urban air mobility, UAM, prosperity. Most of it are electric vertical takeoff and landing, eVTOL, vehicles. For these hovering configurations, the use of contra-rotating propellers allows both small dimension together with sufficient thrust, hence good performance to size compromise. By that, the contra-rotating configuration becomes crucial concerning its performance potential for these vehicles. Unfortunately, performance of contra-rotating configuration is inferior compared to an equivalent single-rotating propeller. Moreover, its acoustic signature is worse, which contradicts the restrictive noise regulation. All of these requires meticulous design of the propulsion system, confronting both its performance and acoustic characteristics. This requires introduction of numerical optimization into the design process, which enables extracting the configuration full potential. Contra-rotating propeller configuration is non-trivial concerning optimal design. This is due to the complex nature of its aerodynamic analysis which implicates also on its noise characteristics. The current effort harnesses several models into unified design framework, which allows optimization of such configuration. The optimization environment is based on ESTECO's modeFRONTIER which incorporates all analysis into the design framework. Using CFD-based analyses is doomed to fail, as part of optimization procedure, due to its very high computer resource requirement. Thus, aerodynamic analysis is based on simple and effective blade-element model, and acoustic prediction is based on standard discretization of the acoustic wave equation together with super-position of the forward and rear propeller contributions. Electric model and motor/propeller matching is based on simple electric relations which are implemented twice for each propeller in the contra-rotating configuration. As part of CRAFT – contra-rotating analysis, fabrication, and test venture, an optimized CR configuration are designed. The paper will present the design process and its outcomes - optimized contra-rotating configuration, with minimal penalty compared to the single-rotating option.

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I. Nomenclature

A	=	propeller disk area
dL	=	blade-element lift
dD	=	blade-element drag
I_0	=	no-load current
I_{ESC}	=	ESC input current
K_v	=	speed constant
$\bar{p}^2$	=	mean square average of the sound pressure
p_{FFT-i}	=	i-th harmonic of pressure wave
Δp	=	perturbation of the static pressure
P_e	=	electric power
P_{shaft}	=	shaft power
r	=	radial coordinate
r_{rel}	=	relative distance
R	=	radius
R_a	=	armature resistance
SPL	=	sound pressure level
$SPL_{Harmonic-i}$	=	i-th harmonic of the sound pressure level
t	=	time
t/c	=	thickness ratio
t_c	=	time duration of a cycle
T	=	thrust
U	=	cross sectional velocity
V_{ESC}	=	input ESC voltage
V	=	velocity
W_a	=	axial induced velocity
$W_{a,1on2}$	=	axial cross-induced velocity, forward propeller on aft
$W_{a,1on2}$	=	axial cross-induced velocity, aft propeller on forward
W_t	=	circumferential induced velocity
$W_{t,1on2}$	=	circumferential cross-induced velocity, forward propeller on aft
Z	=	gap between forward and rear propeller
α	=	angle of attack
β	=	pitch angle
η_e	=	electric efficiency
η_{ESC}	=	ESC efficiency
θ	=	inclination/azimuth angle
ρ	=	air density
φ	=	inflow angle
Ω	=	rotational speed

Acronyms

BEM	=	Blade-element model
BPF	=	Blade passing frequency
CFD	=	Computational fluid dynamics
CR	=	Contra rotating
eVTOL	=	electric vertical takeoff and landing
ESC	=	electronic speed control
FFT	=	Fast Fourier transform
MOGA	=	Multi objective genetic algorithm
NSGA	=	Non-dominated sorting genetic algorithm
OASPL	=	Overall sound pressure level
SPL	=	Sound pressure level
SR	=	Single rotating
UAM	=	urban air mobility

II. Introduction

Contra-rotating propellers were presented from the beginning of flight history with the benefits of improved efficiency and zero torque.¹ The development of drones and VTOL configurations increased the interest in such configuration which led to a flourishing development of analysis tools. Since publication of UBER's elevate² a new interest of this configuration started at the flourishing urban air mobility, UAM, industry. In its basis. Most UAM's are eVTOL (electric vertical takeoff and landing) vehicles. The main source for thrust, especially at hover, comes from rotating-wing such as propellers, rotors, fan etc. During hover, the thrust requirement reaches its peak, i.e. the thrust equal to the vehicle weight. According to simple momentum theory,³ this justified high total disk-area, which improves hover capabilities, mainly decreases the required power to hover. Figure 1 shows a comparison between two of the leading UAM vehicles; on the left the EHang 216 and on the right VoloCity. The two images are scaled equally which emphasizes the smaller footprint of the EHang compared to VoloCity. The EHang uses 16 engines and propellers which are arranged in 8 contra-rotating, CR, propulsion systems. VoloCity consists of 18 single-rotating, SR, propellers which are configured using side-by-side arrangement. This allows the EHang to have span of 5.6 m which is half of the VoloCity (11 m). This implies directly on the vehicle weight, complexity, and price. Note that VoloCity operational takeoff weight is 700 kgf, while the EHang is 600 kgf with the same payload (2 passengers).

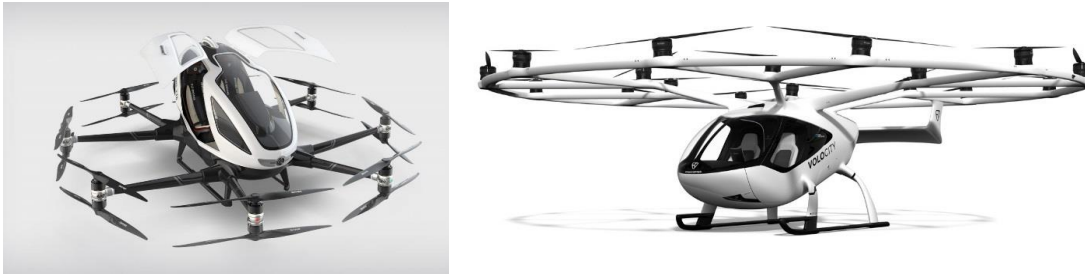


Figure 1, Comparison of EHang and VoloCity (images are scaled equally)
Left: EHang 216⁴, Right: VoloCity⁵

This efficient real-estate exploitation using CR has drawback, on both required power to hover and noise signature. The required power to hover using CR configuration is higher than an equivalent SR. This is demonstrated in Brazinkas et-al paper⁶ which compared the power requirement of CR to SR, for the same thrust requirement. It demonstrated about 20% increase of power for the CR configuration.

Magliozzi⁷ showed that the CR configuration emits higher noise level compared to a SR system which is explained by the interaction noise. Nevertheless, even due to the higher required power, noise signature is expected to increase.

These deficiencies of CR configuration require careful design which can decrease significantly the penalty due to the CR configuration.

Rand and Khromov⁸ used variation-method optimization technique to find the most efficient CR propeller. Their efforts harnessed blade-element model, BEM, similar to the one used in the current effort. Their approach is similar to the classic approach of minimum induced loss which is the basis for optimized SR design method as formulated best by Larrabee⁹. This is less applicable for multi-objective design problems as performance-acoustic tradeoffs. More practical method was presented by Rojas¹⁰ which include a BEM model into numeric optimization design process.

Blade-element model (BEM) is used extensively for rotating wing analysis and design. This includes propellers, open-rotors, wind turbines, axial compressors, fans, etc. The advantage of BEM over many

others elaborate and accurate models, such as CFD based analyses, is the combination of simple model which contains accurate 2-D aerodynamic database for the blade's cross-sections. In that way, BEM is very simple and consumes very low computer resources. The complex effects of viscosity, compressibility, and stall, are introduced through the 2-D aerodynamic database. This is also the main draw-back of BEM – the requirement of valid, high level-of-confidence database, which suites large range of operating conditions, i.e. various Reynolds and Mach number. If such database exists, for axial design condition BEM exhibits excellent match to experimental results.¹¹

This enables researchers to use BEM for open rotor analysis and design due to its benefits. Patrao¹² developed BEM particularly for open rotor design and showed some validation of the method for several propellers and open-rotor from the literature. Extending the BEM for a contra-rotating cases is natural evolvement which answer the need for accurate and efficient contra-rotating wing analysis.

Economon et-al¹³ used only CFD analysis in their design of open rotor. This required careful implementation of Euler equation in an optimization scheme, which although shows the benefits of CFD, emphasized the implementation complexity of CFD in a design process. It can be noted that CFD analysis is still in its initial steps concerning design of rotational wing while BEM probably will maintain its appeal for such a task.

In that scope it is somewhat tempting to use CFD as part of optimization design process. Such an effort was demonstrated by Villar et-al¹⁴ which showed the tradeoff between thrust and efficiency of CR configuration. Although CFD is considered high-fidelity analysis compared to BEM, it has various deficiencies. The main drawback is the high computer resources compared with BEM, thus making BEM much appropriate for practical design by optimization. This drawback is even more pronounced if an acoustic goal is added to the aerodynamic one.

In their research Lepot et-al¹⁵ and Schnell et-al¹⁶ used high fidelity CFD and CFD-based acoustic analyses. They designed CR configuration as part of a joint effort by European consortium using high power computing. Although this approach is appealing, it is far of being practical.

In the current effort a multi-objective design problem is solved using low-fidelity models. These models were developed from previous similar effort for SR configuration. These models were incorporated successfully in SR design process.¹⁸ The resulted SR propellers proved the validity of this process which in the current effort is extended toward CR configuration.

The model which are incorporated in the design framework, share similar simplicity and accuracy levels. Three models consist these analyses: aerodynamics, acoustic, and electric model. In what follows short descriptions of these models together with proper validations, are presented. Then, short description of the design framework and optimization design procedure are depicted, and finally some results are presented.

III. Aerodynamic model

Detailed description of the aerodynamic model is given in Refs. 20 and 21. These include also rigorous validation of the model through comparison to test results which are available in the literature. Still, short description is given in the following paragraph, followed with relevant validation.

Left part of Figure 2 shows a single-rotating propeller with an axial free stream velocity, V . In this figure the propeller equivalent actuator-disk model is also presented. The propeller is replaced with an infinitesimal thin actuator-disk which introduces axial and circumferential momentum to the flow. The additional momentum is resulted with axial and circumferential induced velocity components, W_a and W_t , respectively. The source of these induced velocities is the propeller thrust and torque. According to Newton's third law, the thrust and torque that acts on the propeller is reacted by the same thrust and torque which acts on the flow. Thus, the air is accelerated to the axial direction due to the thrust and the wake is swirled due to the torque, i.e. axial and circumferential induced velocities are resulted.

In the current effort, the SR model is extended to a contra-rotating case as appear in right part of Figure 2. It shows a contra-rotating propeller and its equivalent actuator-disks model. Each of the two actuator-disks acts the same manner as a single-rotating case. Main difference is the cross-induced velocities. Except of the axial and circumferential induced velocities which the disk induces on itself, the disks induce velocities on each other. The rear disk induces axial cross-induced velocity on the forward disk, $W_{a,1on2}$, and the forward disk induces on the rear disk both axial cross-induced velocity, $W_{a,1on2}$, and circumferential cross-induced velocity, $W_{t,1on2}$. Note that the rear disk does not induce circumferential induced velocity on the forward disk – only the wake swirls.

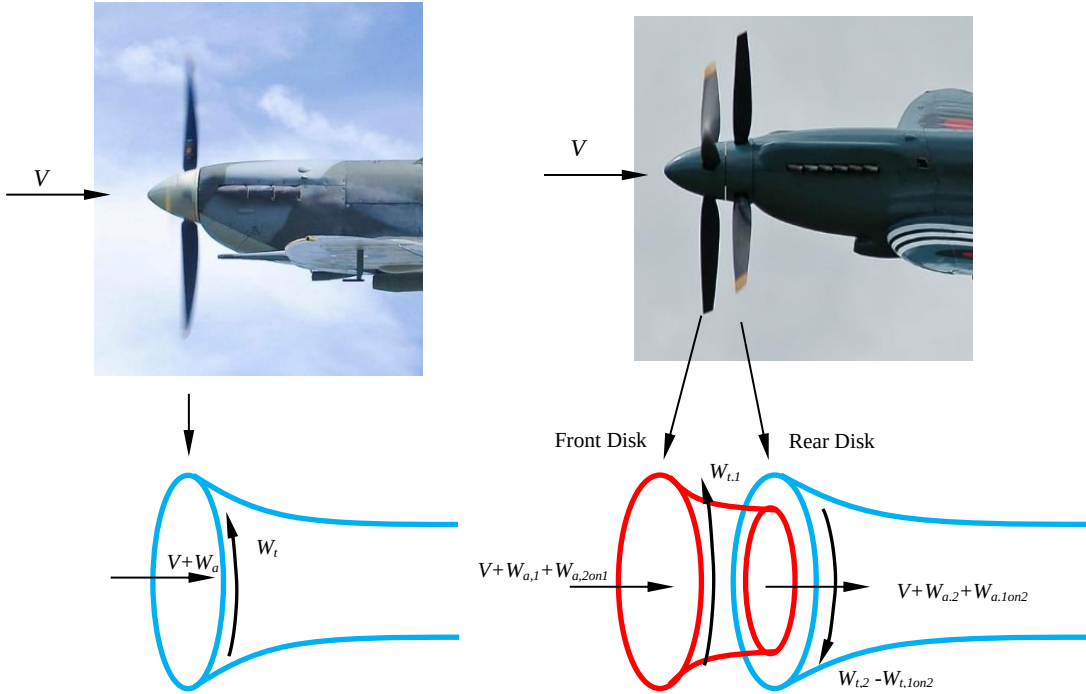


Figure 2, Single- (left) and Contra- (right) rotating propeller schematic of the actuator disk depicting the velocity components. Supermarine Spitfire's figures are taken from Ref. 19

Figure 3 shows a typical blade-element cross section. The rotating propeller's blade is sliced to small blade-elements. The velocity components on the blade sections are summed resulting with total velocity, U , and local angle of attack, α . The blade-element acts as a 2-D wing section, thus it produces lift and drag forces, dL and dD , respectively. Summing all blade-element lift and drag components is resulted with the total thrust and torque which acts on the propeller. The summation of axial and circumferential induced velocity, ΣW_a and ΣW_t , respectively, impacts the velocity diagram, thus influence U and α . Through these, each propeller (forward and aft) can be independently calculated. Note that the 3 cross-induced velocities, $W_{a,1on2}$, $W_{a,2on1}$, and $W_{t,1on2}$, are required. Thus, an iterative scheme finds these components.

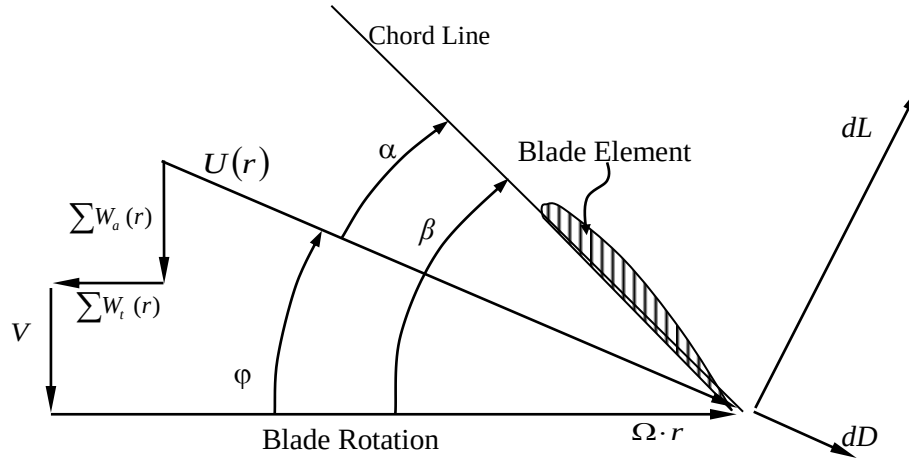


Figure 3, Blade cross-section with various velocity components

Although various validations were conducted versus results from the literature, more meticulous validation is required for hovering case, which is a unique $V=0$ case. As a validation case, an off-the-shelf propeller, Mejzlik 18x6 is used. Figure 4 depicts the installation of two Mejzlik 18x6 propellers on the CR test rig. This rig allows the change of propeller gap, Z , and it is installed with internal balance. Hence, during test all data can be logged such as: torque, speed, thrust, and electric voltage and current. This is done for each engine separately. In addition, the test rig is portable; thus it can be transferred easily to an anechoic chamber or wind tunnel.

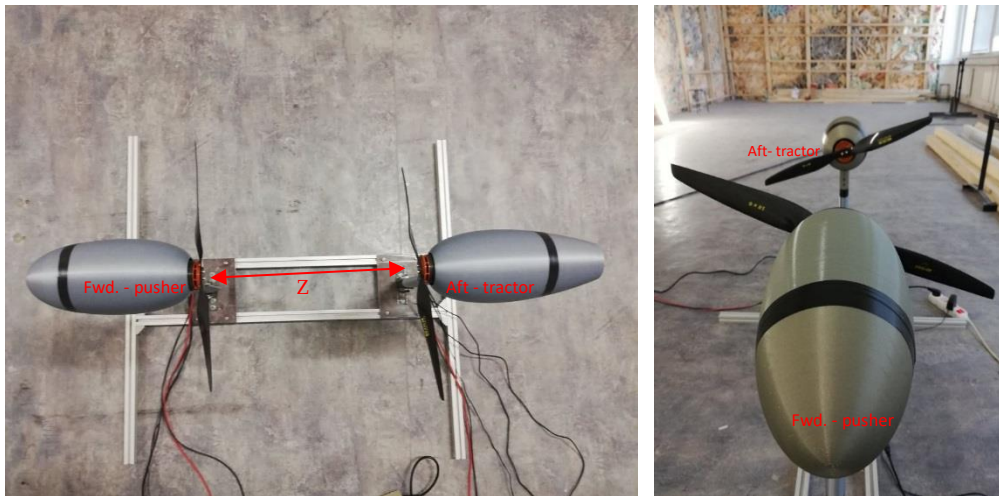


Figure 4, Mejzlik 18x6 propeller on the CR test rig

Figure 5 shows the test results of the CR configuration, based on the reference Mejzlik 18x6 propellers. The results contain thrust and shaft power, thus it is not related to the electric system, only the propellers aerodynamic. The test was done for 3 propeller gaps, $Z = 0.1, 0.3, 0.5$ m and various propeller speeds combinations. The trend lines of the test and analysis fit well. Average thrust error (for the total thrust) is 6.3% and the total power average error is 3.4%. This exhibits the model accuracy and validates it.

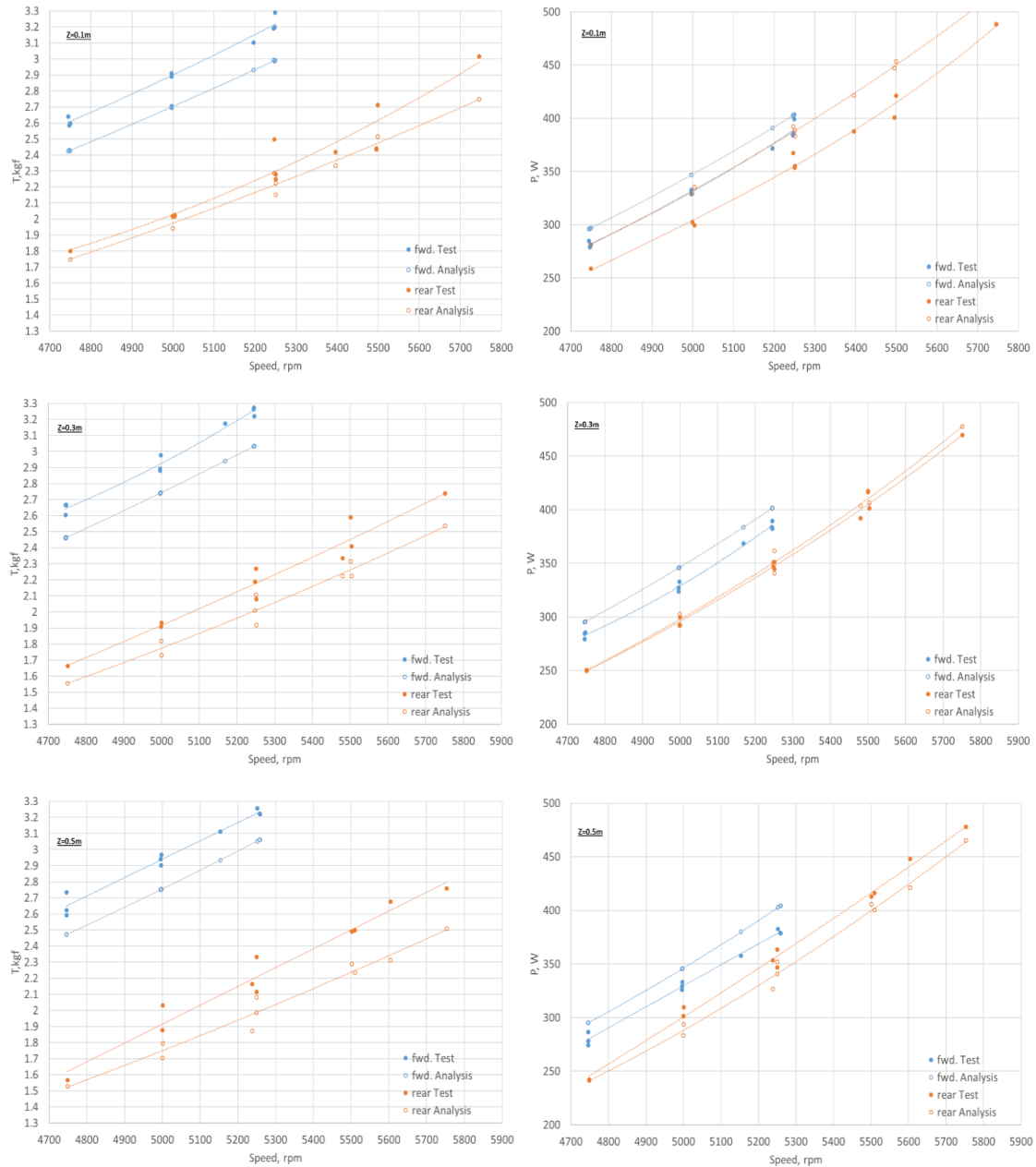


Figure 5, CR performance test results vs. analysis
Left Column: Thrust, Right column: Shaft Power

IV. Acoustic Model

The acoustic model in the current effort is based on the same model used in Ref. 22, which is shortly described in what follows. Note that the model was developed originally for a single rotating propeller.²³ It is based on a solution of the constrained wave equation, known as the Ffowcs-Williams/Hawkings equation.²⁴ This equation contains three forcing terms: thickness, loading, and vortex, which are represented mathematically by monopole, dipole, and quadrupole noise emitting sources, respectively.²⁵ Hanson and Fink²⁶ showed that for thin blades with cross-sections operating in subsonic or transonic conditions, the vortex (quadrupole) term can be neglected. Thus the solution of the Ffowcs-Williams/Hawkings equation includes two terms - the thickness noise, Δp_{Thick} , and the loading noise, Δp_{Load} :

$$\Delta p(\mathbf{x}, t) = \Delta p_{Thick}(\mathbf{x}, t) + \Delta p_{Load}(\mathbf{x}, t) \quad (1)$$

The solution of the Ffowcs-Williams/Hawkings equation is obtained after applying Green's function.²⁴ Then the solution is discretized³⁰ and the following approximations are adopted:

- a. The velocity of the noise source is sub-sonic.
- b. The noise source is much faster than the observer.
- c. The propeller's rotational speed and the velocity of the air vehicle do not vary with time.

The final expressions for the loading and thickness noise are presented thoroughly in Ref. 31. Note that the assumption of subsonic noise source limits the current analysis for subsonic propeller tips, which in some prop-fans limits the operating conditions to be analyzed. Moreover, high transonic blades, which reach very close to unity Mach number, might exhibit poor analyses results.

The loading noise term is comprised of two components: the far field and near field loading noise. These two terms differ by the power of the relative distance between the observer and the noise source, r_{rel} . While the far field term is proportional to $1/r_{rel}$, the near field term is proportional to $1/r_{rel}^2$, thus it becomes relatively small as r_{rel} increases.

Calculation of the noise produced by a propeller is based on dividing the propeller's blades into small elements (similar to the aerodynamic blade-element model, BEM). The pressure at a certain point in space, at a certain time, is obtained by summing up the contributions of all elements. Then the same process is repeated for the next time step, where the new positions of the air-vehicle and propeller blades, at that time, are considered. The pressure wave, $\Delta p(x, t)$, is defined after repeating this process for a complete cycle, e.g. for a two-bladed propeller, one noise cycle is equivalent to half a revolution.

If the time dependent pressure wave, $\Delta p(t)$, is known, its spectral analysis defines the Sound Pressure Level (SPL) and its different harmonics. The mean square average of the sound pressure, $\bar{p}^2$, and the resultant sound pressure level, SPL (given in [dB]), are defined as follows:

$$\bar{p}^2 = \frac{1}{t_c} \int_0^{t_c} \Delta p^2(\mathbf{x}, t) \quad (2)$$

$$SPL[dB] = 10 \cdot \log_{10} \frac{\bar{p}^2}{p_{ref}^2} \quad (3)$$

where t_c is the time duration of a noise cycle. The reference pressure for air is usually chosen as $p_{ref} = 20 \cdot 10^{-6} [\text{Pa}]$. Using Fast-Fourier-Transform (FFT) the different harmonics of the pressure wave, p_{FFT-i} , and the sound pressure harmonics, $SPL_{Harmonic-i}$, are calculated:

$$\bar{p}^2 = \sum_i \frac{1}{2} \cdot p_{FFT-i}^2 \quad (4)$$

$$SPL_{Harmonic-i}[dB] = 10 \cdot \log_{10} \left(\frac{p_{FFT-i}}{p_{ref}} \right)^2 \quad (5)$$

As stated above, this procedure was used successfully for a single-rotating propeller^{17,23}. The validation shows good match for single-rotating propellers, thus the potential for a contra-rotating case will be shown in what follows.

The acoustic signature of the contra-rotating configuration is treated as the superposition of two single rotating cases. It is well known that contra-rotating configuration can't use a simple superposition, due to the interaction noise levels.^{27,28} Still, in some cases the superposition gives fair agreement both for the absolute sound pressure levels and to its trends and characteristics.

Considering two isolated single-rotating propellers sound-pressure-levels, *i.e.* SPL_1 and SPL_2 . Each represents a square pressure term, $\bar{p}_1^2$ and $\bar{p}_2^2$, respectively. The superposition squared average pressure is $(\bar{p}_1 + \bar{p}_2)^2$, thus the super-positioned sound pressure level, SPL_{1+2} is defined in Eq. (6).

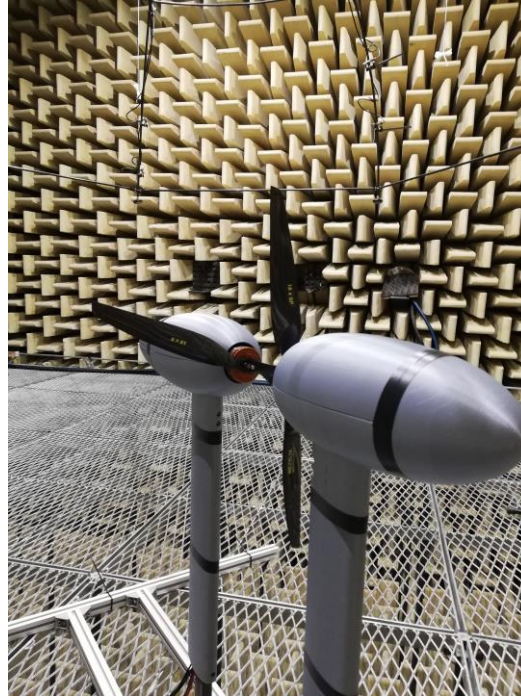
$$SPL_{1+2}[dB] = 10 \cdot \log_{10} \frac{(\bar{p}_1 + \bar{p}_2)^2}{p_{ref}^2} \quad (6)$$

Using the definition in Eq. (3), one can substitute SPL_1 and SPL_2 to (6) as depicted in Eq. (7).

$$SPL_{1+2}[dB] = 10 \cdot \log_{10} \left(10^{\frac{SPL_1}{20}} + 10^{\frac{SPL_2}{20}} \right)^2 \quad (7)$$

Eq. (7) enables a very simple estimation of a contra-rotating configuration. First the contra rotating loading is estimated. This loading is based on a BEM aerodynamic analysis which includes forward and rear propeller interaction. Thus, the aerodynamic properties of the contra-rotating configuration are calculated in a proper manner, rather super-positioned. Then, the acoustic signature of the forward and rear propeller is calculated separately. Each resulted with the total SPL and the harmonic $SPL_{Harmonic-i}$. Finally, using Eq. (7), the super-positioned total SPL and the super-positioned harmonic $SPL_{Harmonic-i}$ are calculated.

The acoustic analysis was validated through a test held in Resideo 5×5×5 m anechoic chamber, located in Brno, Czech Republic. Figure 6 shows the faired test rig with the Mejzlik 18×6 propeller installed in the chamber. A total of 20 microphones on a standard hemisphere²⁹ was hung above the propeller - it can be noticed in Figure 6. The propeller was located off-center from the array and at a 45° angle so as to minimize the number of microphones which would be located in the wake or recirculation flows. As a result, a range of radii from 1.1-2.9 meters and propeller azimuthal angles $\theta = 75.2 \div 154.7^\circ$ was recorded. Note that $\theta = 90^\circ$ is in the propeller disk plane, while higher angles refer to the wake region.



**Figure 6. Mejzlik 18×6 CR propellers installed in Resideo anechoic chamber
Part of the microphone dome can be noticed above the test rig**

The results for 3 of the microphones, located at different azimuth angles, θ , are presented on Figure 7. The results from the test are tonal spectra integrated from narrow-band autospectral densities ($\Delta f = 5.9$ Hz) about each of the tonal peaks. This method varies slightly from method of Farassat³⁰ which uses a Δf equal to the BPF, blade pass frequency. Note that the microphones are located at different distances from the propeller, thus the comparison between microphones SPL 's is useless without a distance correction.

All results exhibit deteriorated model accuracy for high azimuth angles- rear location. This can be seen in Figure 7. The front ($\theta=88^\circ$) microphone exhibits high noise level, thus the tonal test results are capable of locating 3 harmonics. The $\theta=115^\circ$ is capable of locating 2 harmonics and the most rear microphone at $\theta=150^\circ$ measures only 1. As the SPL is below 55 dB, the tonal harmonics are hard to decipher out of the microphones measurements. This behavior is not typical for propellers in a free field and is related to the test environment and test data reduction. Similar results were observed in the work of Block.³² Block mentions enforcement to this hypothesis:

Typically, the results from Method 1 and Method 2 agree for the first few harmonics, which are above the background noise of the tunnel. As the harmonic number increases, Method 2 gives the levels of the background noise, whereas Method 1 gives levels below the background noise which appear to follow the trends expected for propeller noise.

In the above quote “method 1” refers to normal behavior of propeller SPL harmonics, while “method 2” refers to results similar to what presented here. Similar methods with the same phenomena are discussed in Ref. 33. There, several post-processing techniques are implemented to better recognize and separate the tonal and broadband noise.

A recent work³⁴, has shown these increased peaks to be resultant from unsteadiness in the flow caused by recirculation in a closed environment. In Ref. 34 the authors show the importance of air recirculation in the test chamber which can cause some of the high frequency deviation. In the current effort the knowledge gained from this reference was applied to the current data set.

On top of the measurement and deciphering issues, the rear position analysis results suffer from the lack of interaction noise model. This was discussed in Ref. 22 which exhibits comparison of the current model to other results from the open literature. In all cases, the wake’s noise prediction is poor, especially in CR case which include high levels of interaction noise components.

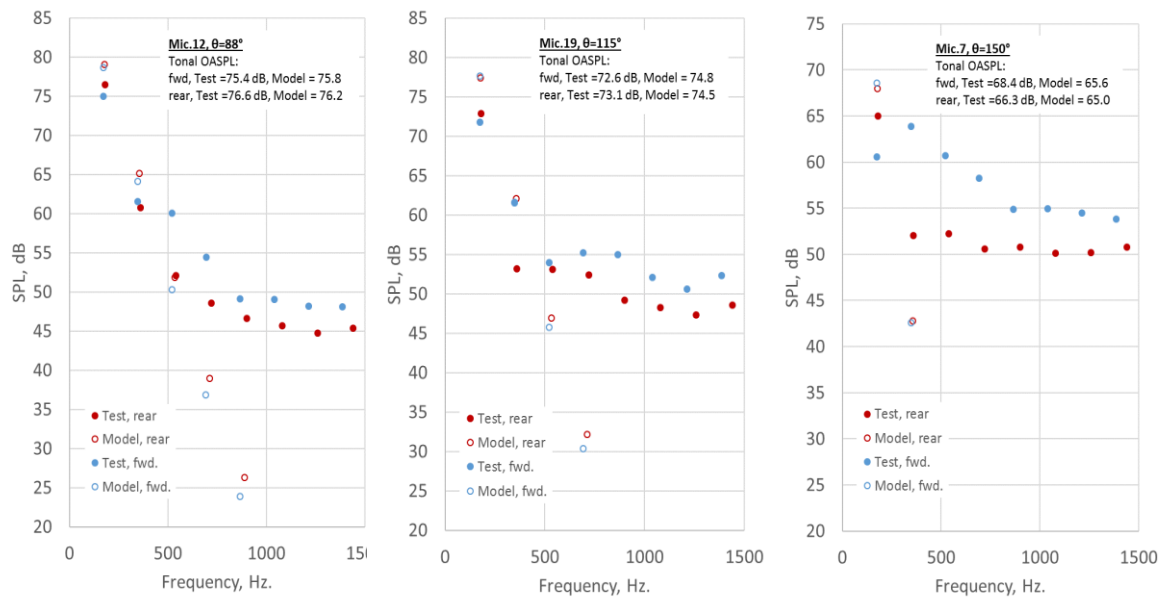


Figure 7, Acoustic results for reference CR (Mejzlik 18x6), Test vs. Analysis SPL, Z=0.1m

Figure 8 shows the residuals (test minus model prediction) for all test results. The charts include model errors for both the 1st harmonic and tonal-OASPL. First harmonic residual reaches to 5dB. This seems high, but the trends seem accurate and consistent. All residuals are contained in band of about 2~3dB (for low azimuth $\theta<120^\circ$). This makes the residuals to be consistent, for both microphone locations, forward and rear propeller difference, and propeller gap, Z. This consistency fits well the results presented in Ref. 22

which also showed comparison to tests (from the literature) with highly consistent agreement of trends, although somewhat poor absolute agreement to results.

Note that in design, the most important indicator is the trend accuracy rather absolute accuracy. The design process, especially design by optimization, deals with improvements compared to a reference case. Using physical, validated, and trend-accurate models, assures the success of the design procedure.

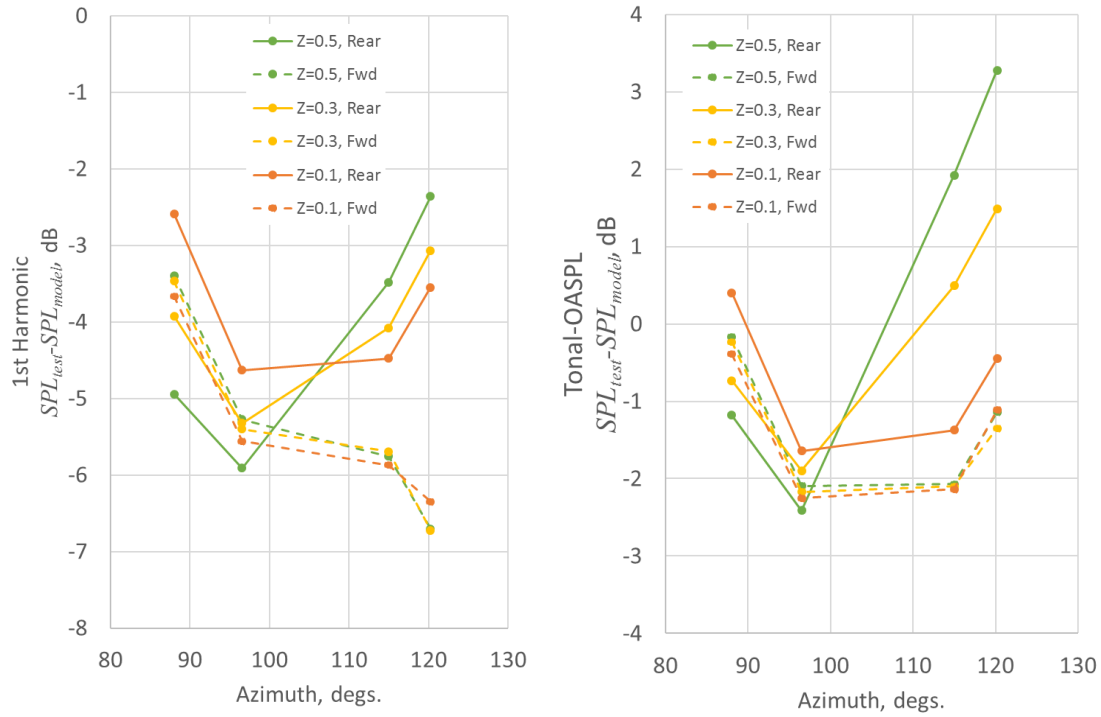


Figure 8, CR acoustic test results vs. analysis
Left: 1st harmonic, Right: total tonal-OASPL

V. Electric Model

In the current effort a simple motor model is used to find the required electric power. The model is based on four constant parameters: speed constant, K_v , armature resistance, R_a , no-load current, I_0 , and controller efficiency, η_{ESC} .³⁵ The model is based on the following assumptions:

- Power factor is equal to unit. This assumption is applicable to small brushless Permanent Magnet (PM) motors.
- Magnetic losses (eddy/Foucault Current and magnetic hysteresis) can be neglected.

Using the above assumption allows to estimate ESC electronic system control, output current, I_{ESC} , and voltage, V_{ESC} , through Eq. (8).

$$\begin{aligned} I_{ESC} &= \frac{P_{shaft} \cdot K_v}{\Omega} + I_0 \\ V_{ESC} &= \frac{\Omega}{K_v} + I_{ESC} \cdot R_a \end{aligned} \quad (8)$$

Total electric power, P_e , used from the energy storage (batteries), is calculated using the ESC efficiency, η_{ESC} , as depicted in Eq. (9).

$$P_e = \frac{I_{ESC} \cdot V_{ESC}}{\eta_{ESC}} \quad (9)$$

Thus, the electric system efficiency, η_e , can be calculated using the ratio of P_{shaft} and P_e

$$\eta_e = \frac{P_{shaft}}{P_e} \quad (10)$$

The motor manufacturer suggested some values for the above parameters (K_v , R_a , I_0 , and η_{ESC}). Still, an optimized estimation was done to find the parameters values which give best agreement to the performance test. To increase the level of accuracy, the speed constant, K_v , is assumed to be linear dependent with the torque, Q , of the motor.

The test rig (Figure 4) is consisted from two separate, identical electric systems. Each is based on Sobek 20-38 Spider AC-motor with Kontrol-X 55LV electronic speed controller, ESC, as shown in

Figure 10. This rig has been used for vast number of propellers, SR and CR configurations, with the same propulsion system. Using all vast gathered data, an electric system model (namely the system parameters) was substantiated. Note that for the electric model only the input current and voltage is required, and the output speed and shaft power. This makes the model indifferent to the installed propeller characteristics.

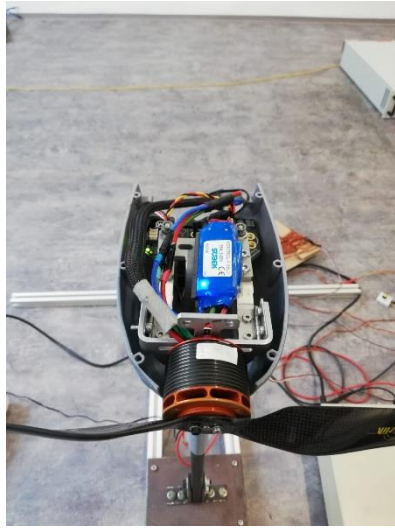


Figure 9, Test rig and electric propulsion system buildup

Comparison of the model to the measurements is presented in Figure 10. The residuals between model and test are presented for the electric power (left) and electric efficiency (right). Both exhibit accurate results with average error of $\mu=7.9\text{W}$ and 0.75% , respectively. Note that this agreement was achieved after optimizing the model parameters. Still, it covers various propeller and conditions, thus validates the electric model which is used as part of the design process.

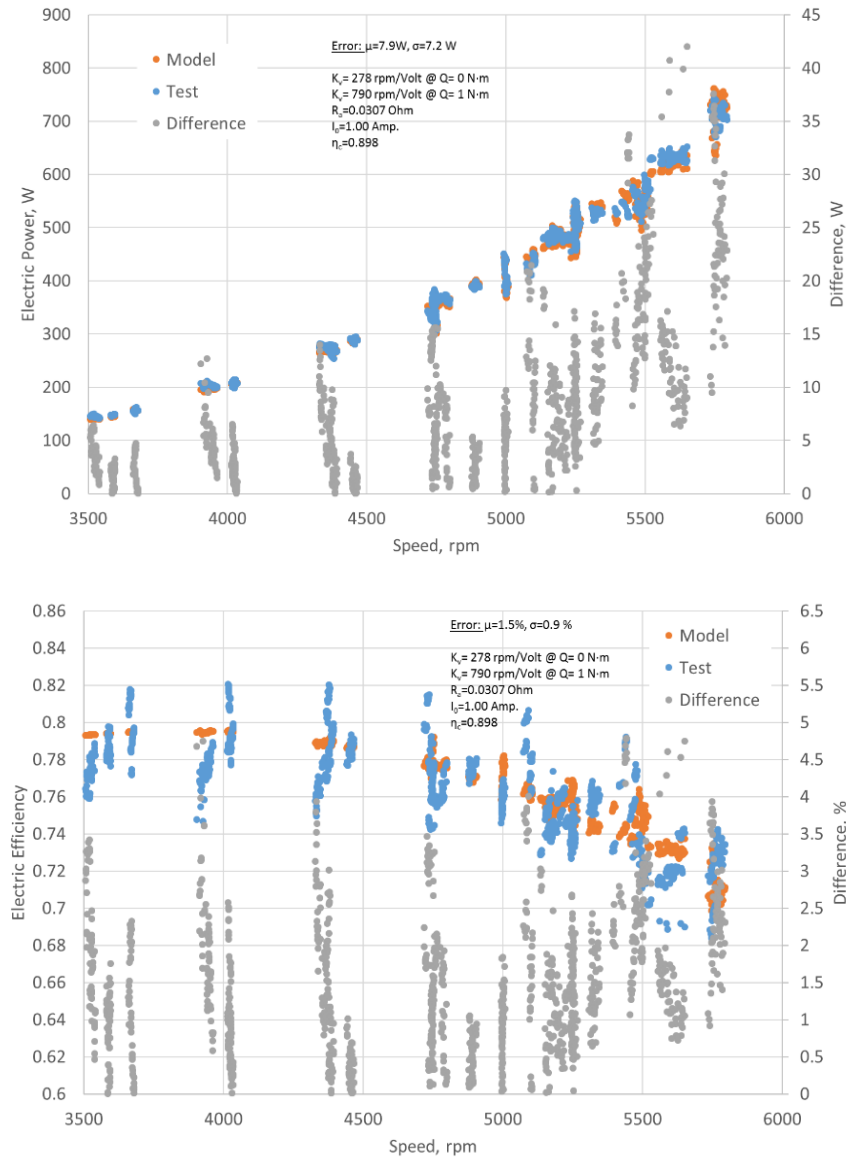


Figure 10, Electric system results – test versus optimized model
 Upper chart: required electric power, P_e , Lower Chart: electric system efficiency, η_e

VI. Design procedure and framework

Design procedure is similar to former cases accomplished with the same tools. The analysis tools (BEM, acoustic model, and electric model) are integrated into ESTECO modeFRONTIER framework.^{17,36}

Figure 11 presents a screen capture of the modeFRONTIER framework. modeFRONTIER design environment enables the integration of different simulation models into a single synergetic design tool. In addition, it allows the use of various optimization procedures, thus a MDO tool is obtained. In the current case, first the propeller performance is calculated and then the propeller acoustic is estimated. The use of modeFRONTIER enables an easy usage of any of the input or output parameters, either as design variables or to include them in the goal function and constraints.

The framework (Figure 11) starts with geometry parameterization of the blade properties distribution (pitch, chord and thickness), for each of the propellers. Geometric pre-analyses are implemented twice, for the forward and rear propeller, using Excel. The geometry pre-analysis is used to parametrize the design variables which are the pitch, β , thickness-ratio, t/c , and chord-to-radius ratio, c/R , distribution along the blade. The current effort uses a Bezier spline to achieve smooth distribution of the geometry, 5 parameters for each distribution at each propeller (forward and aft) has its own design. Thus, the design problem contains total of 30 design variables which are related to the blade's geometry. All airfoils are based on the Mejzlik 18×6 cross sections and the propeller radius is $R=0.23$ m.

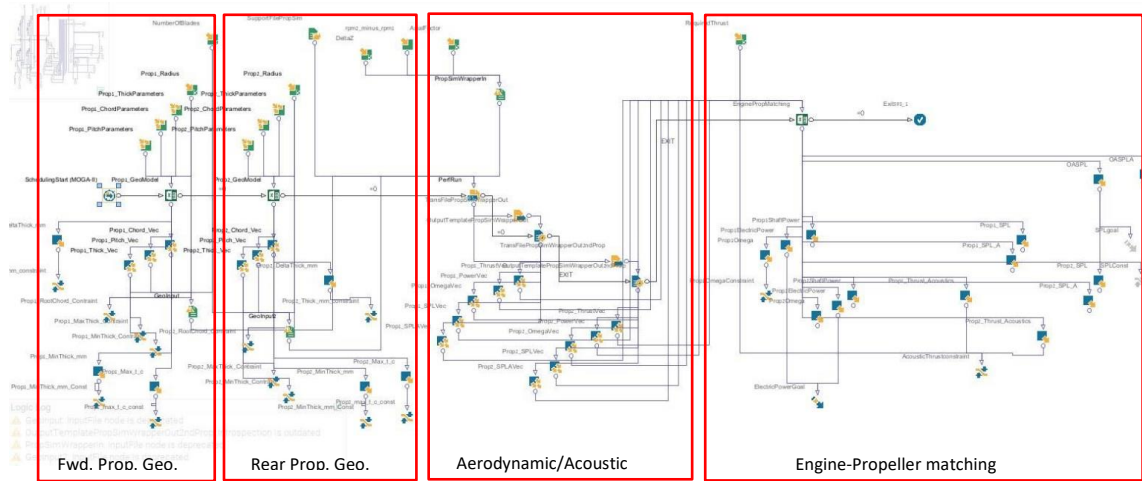


Figure 11, modeFRONTIER design framework screen capture.

To ensure the structural properties of the optimized designs, two geometric constraints are to be satisfied. First the blade thickness distribution should not be lower than the original Mejzlik 18×6. Thin blade might be “soft” or exposed to high stresses, which might cause unacceptable aeroelastic behavior, and high deflections. In addition, the root chord should not be larger than the Mejzlik 18×6’s root chord. This might cause a very thick hub which increases the propeller weight.

To overcome these issues, the design procedure incorporated two geometric constraints over the thickness distribution and root chord. The first constraint limits the thickness distribution and the second the root chord. The thickness distribution, t (rather t/c) has to be higher or equal to the Mejzlik 18×6 up to $r/R=0.90$, with a tolerance of 0.1 mm. The blade tip ($0.9 < r/R < 1$) was freed from this constraint – the impact over the design was high and it seems the importance of this constraint, at the very tip of the blade, is less important. The root chord is limited to $c/R < 0.15$. Note that the Mejzlik 18×6’s $c/R = 0.14$ at the root, thus small increase of the root chord is allowed.

After the geometric definition of the forward and rear propeller aerodynamic and acoustic analyses are conducted, for different values of speed, Ω . The results are used as input to the engine/propeller matching scheme which is implemented using Excel. This matching finds the correct speed, Ω , which the total CR system thrust reaches to 5.6 kgf – this is considered as the design thrust goal. An additional design variable

is the speed difference between the front and rear propeller. This enable the flexibility of the scheme in finding an optimized design with different propeller speeds. This parameter increases the total number of design variables to 31.

The entire analysis procedure is done repeatedly by the optimizer, until the proper design variables are found. Two design goals are defined. The first is the total electric power, P_e , of the system (for thrust of 5.6 kgf), while the second goal is the tonal-OASPL at azimuth of $\theta=100$ deg. This makes the optimization problem a multi-objective one, thus Pareto frontier is found as a compromise between the performance and acoustic goals.

This procedure is repeated for each propeller gap, Z , and was accomplished, in the current effort, for $Z = 0.1, 0.3$, and 0.5 m. At each case, two single-objective optimization problems are solved. Once the minimum electric power goal is used – the minimum P_e propellers are found. Then, the minimum OASPL goal is used, thus the minimum OASPL propellers are found. Only then, the Pareto frontier is found using multi-objective schemes such as MOGA (multi objective genetic algorithm) or NSGA (non-dominated sorting genetic algorithm). This hybrid, multi-phase optimization procedure is discussed thoroughly in Ref. 37.

VII. Design Results

As discussed above, two separate single objective optimizations are conducted, prior to the multi-objective, tradeoff design process- minimum P_e and minimum OASPL. Then the results are used as DOE for the multi-objective tradeoff design problem.

A. Single objective minimum P_e

The single objective minimum P_e optimization results are presented in Figure 12. The chart also contains the reference Mjzlik 18x6 in 4 different configurations:

- 2 blades, SR
- 4 blades, SR
- 2 SR, 2 blades each, side-by-side
- 2 blades each, CR

For the minimum P_e case, the best configuration is 2 separate (side by side) SR propellers. The second-to-best option is the CR configuration. According to the optimization results, the reference CR configuration is improved by 7%, decreasing substantially the penalty over the SR side-by-side. Numeric results are presented in Table 1.

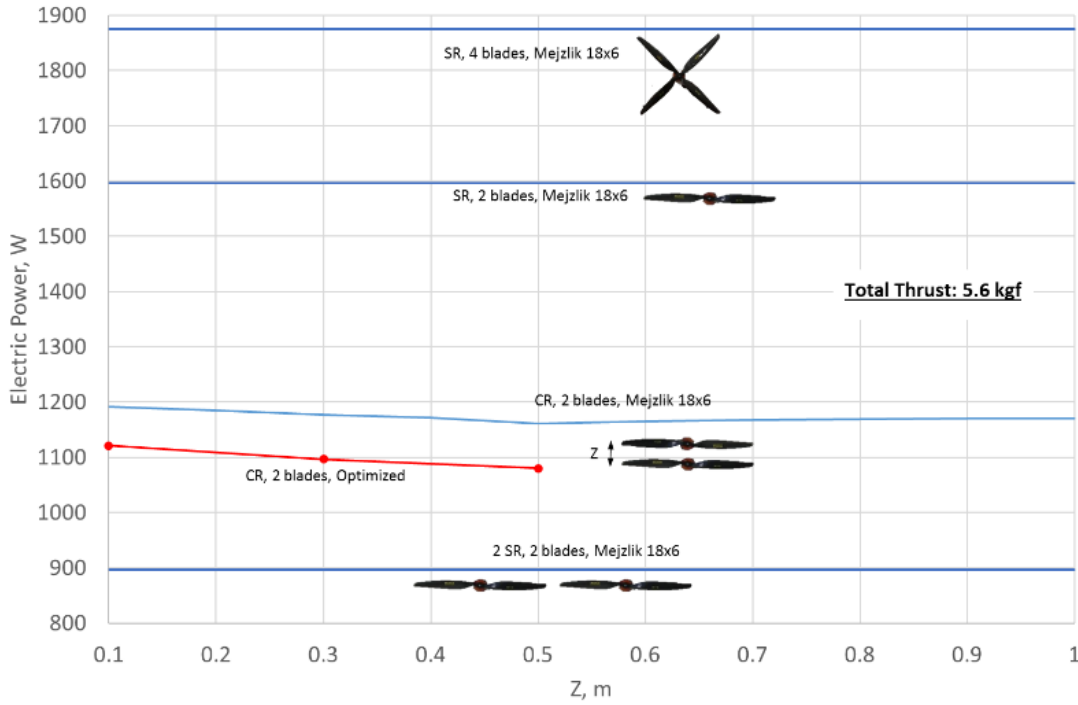


Figure 12, Design results for min. P_e single-objective

From the results the main mechanism of the optimized design is revealed, i.e. rotational speed is increased. This improves the electric system efficiency, η_e . This is probably the reason that the forward and rear propeller share the same speed, namely $\Delta\Omega=0$. The optimization “pushes” both propellers toward better electric efficiency. This can be seen in the second part of Table 1 which depicts each propeller performance (thrust, electric power, shaft power, figure-of-merit, and electric efficiency). Figure-of-merit represents the aerodynamic capability of the hovering propeller and it is defined in Eq. (11).

$$FM = \frac{T^{1.5}}{P_{shaft} \sqrt{2 \cdot \rho \cdot A}} \quad (11)$$

where T is the thrust, ρ is the air density, and A is the propeller disk area.

Table 1 reveals that the optimized propeller FM decreases to allow the increase of η_e . This emphasizes the importance of the electric system over the propulsion system.

The propellers thrust distribution tend to increase the thrust produced by the forward propeller over the rear, which is due to the much higher figure of merit of the forward propeller - it is worthwhile to use higher FM propeller rather the low FM propeller.

One should remember that real-world restriction will include maximal required thrust (peak). Due to an increased hover rpm might reduce that peak thrust. Thus, for real-case scenarios, using the min. P_e propeller might cause some issues for the peak thrust.

Table 1, min P_e , Comparison of CR propeller systems
Total thrust = 5.6 kgf

		Total P_e , W	OASPL, dB	(Fwd.) Prop. Speed, rpm	$\Delta\Omega$, rpm (Rear-Fwd.)
Z=0.1m	Mejzlik 18×6	1,190	74.9	5,500	200
	Optimized 18”	1,120	76.5	6,350	0
Z=0.3m	Mejzlik 18×6	1,170	75.2	5,350	550
	Optimized 18”	1,100	76.2	6,200	50
Z=0.5m	Mejzlik 18×6	1,160	75.1	5,350	600
	Optimized 18”	1,080	76.6	6,400	0

		Thrust, N		P_e , W		P_{shaft} , W		FM		η_e	
		Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear
Z=0.1m	Mejzlik 18×6	31.4	23.5	583	608	430	450	0.658	0.409	0.737	0.740
	Optimized 18”	31.9	23.0	562	560	437	435	0.664	0.408	0.777	0.778
Z=0.3m	Mejzlik 18×6	30.3	24.6	536	641	400	475	0.673	0.413	0.746	0.741
	Optimized 18”	33.0	21.9	579	516	445	405	0.686	0.407	0.769	0.785
Z=0.5m	Mejzlik 18×6	30.1	24.8	526	635	393	472	0.677	0.421	0.748	0.743
	Optimized 18”	33.4	21.5	580	499	450	396	0.690	0.407	0.775	0.793

The optimized propeller geometry for minimum P_e is given in Figure 13 with the distribution of the pitch, chord, and thickness ratio along the blade radii. All gaps, Z , exhibit very similar geometry which shows the low influence of Z over the optimized design.

For all designs, there is a distinctive difference between the forward and the rear propeller. The forward is very similar to the reference Mejzlik 18x6 with somewhat lower pitch. This allows the higher rotational speed which lead the design trend for minimum P_e . The rear exhibits much lower chord with increased pitch. This is due to the lower thrust requirement; thus the propeller solidity can be decreased and the. The lower chord distribution requires higher thickness-ratio. This is done to comply with the geometric thickness constraint.

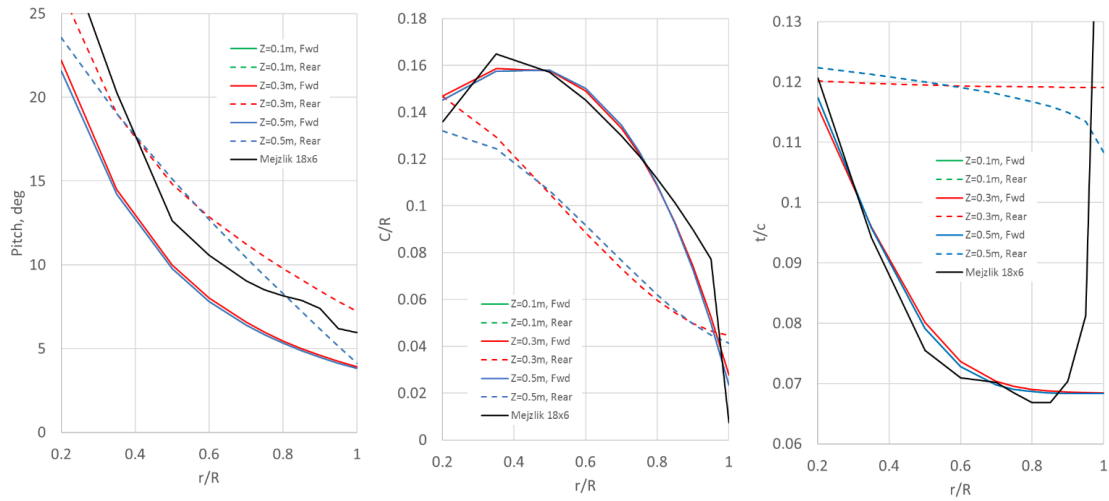


Figure 13, min P_e , Optimized CR propellers' geometry

B. Single objective minimum OASPL

The second single-objective case is the minimum OASPL which is depicted in Figure 14. As before, some reference configurations, based on the Mejzlik 18x6 are presented as well. The best configuration for minimized tonal-OASPL is the 4 bladed SR propeller. Increased number of blade is an immediate way of decreasing tonal noise which was demonstrated in Ref. 18. The two separate, side-by-side SR propellers exhibit substantially higher noise levels but lower than the Mejzlik 18x6 based CR configuration. By optimizing, the tonal-OASPL level of the CR configuration can reach the 2 separate SR Mejzlik 18x6 propellers. Thus, diminish the noise penalty of the CR configuration. Still, as shown in Table 2 this comes with large performance price.

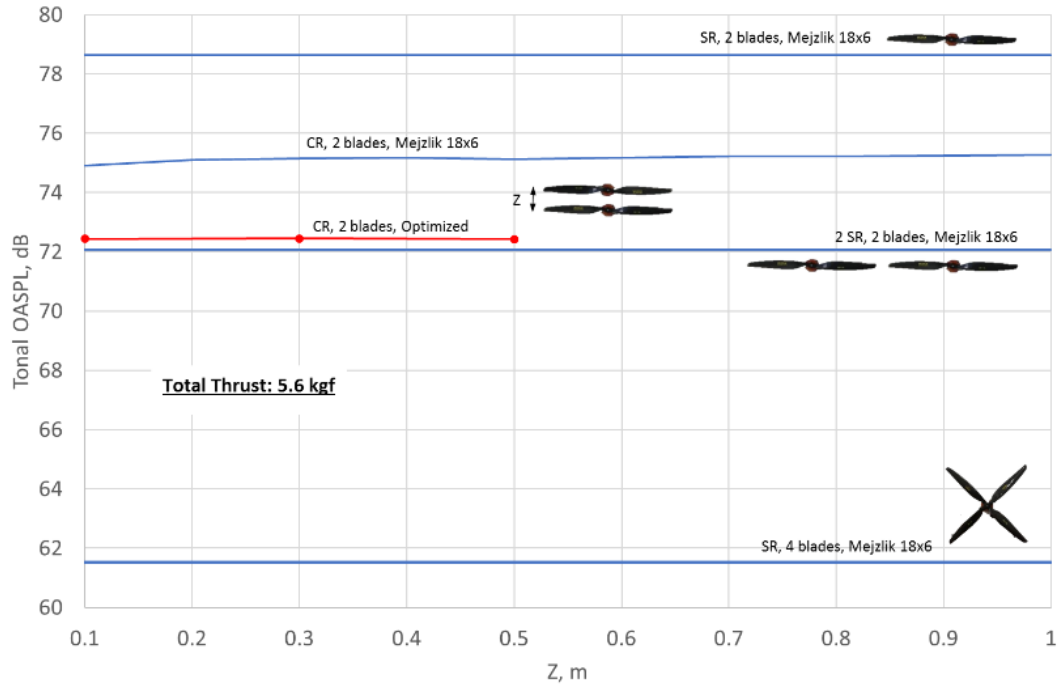


Figure 14, Design results for minimum OASPL single-objective

Table 2 compares the reference 18x6 CR to the optimized CR. Main mechanism of reduced noise is decreased speed. This is evident in the propeller geometry which exhibits high pitch compared to the Mejzlik 18x6 (Figure 15). Table 2 exhibits the performance scarification for the reduced speed. Both shaft and electric power are increased which is also appear in the decreased figure-of-merit and electric efficiency.

Similar to the min P_e results, the differences between the various Z geometries is negligible. The forward propeller is similar to the reference Mejzlik 18x6 with higher pitch. This enables the decrease of propeller speed. The rear propeller exhibits much lower chord; hence higher thickness ratio i.e. maintains the thickness constraint. The rear propeller also exhibits much higher chord which allows it to decrease its speed and produced enough thrust.

Table 2, min SPL, Comparison of CR propeller systems
Total thrust = 5.6 kgf

		Total P_e , W	OASPL, dB	Prop. Speed (Fwd.) rpm	$\Delta\Omega$, rpm (Rear-Fwd.)
Z=0.1m	Mejzlik 18x6	1,200	74.9	5,400	350
	Optimized 18"	1,400	72.4	4,500	150
Z=0.3m	Mejzlik 18x6	1,200	75.1	5,250	700
	Optimized 18"	1,350	72.4	4,550	100
Z=0.5m	Mejzlik 18x6	1,150	75.1	5,250	750
	Optimized 18"	1,300	72.4	4,650	0

		Thrust, N		P_e , W		P_{shaft} , W		FM		η_e	
		Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear
Z=0.1m	Mejzlik 18x6	30.3	24.6	552	641	410	470	0.656	0.418	0.743	0.733
	Optimized 18"	29.9	25.0	656	739	428	470	0.617	0.428	0.652	0.636
Z=0.3m	Mejzlik 18x6	29.2	25.7	504	675	379	496	0.672	0.423	0.751	0.734
	Optimized 18"	30.6	24.3	660	703	431	457	0.632	0.423	0.652	0.650
Z=0.5m	Mejzlik 18x6	29.0	25.9	495	669	372	493	0.676	0.430	0.753	0.737
	Optimized 18"	31.7	23.2	690	632	449	424	0.640	0.424	0.650	0.671

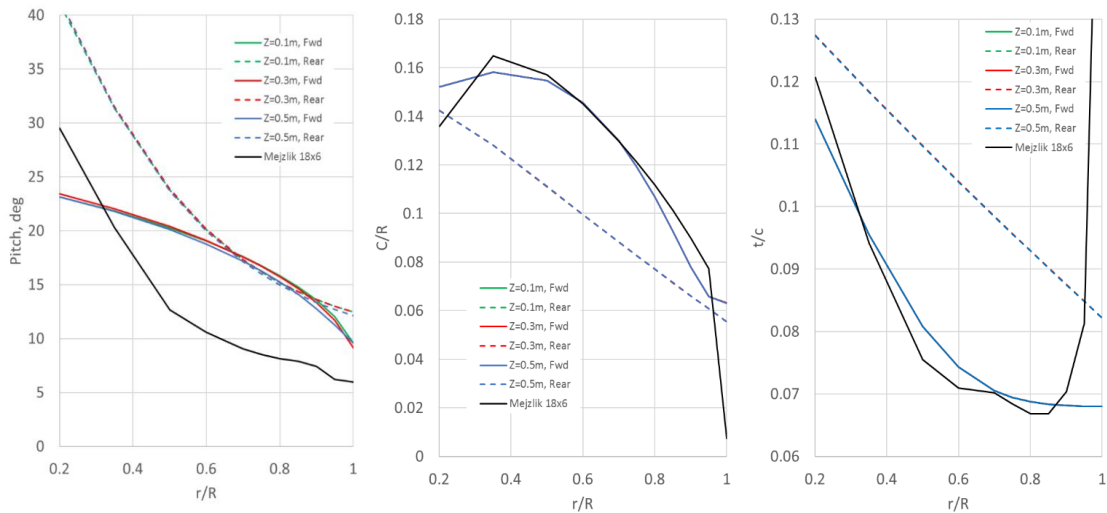


Figure 15, min OASPL, Optimized CR propellers' geometry

C. Multi-objective results

The tradeoff between performance and acoustic was found using multi-objective optimization with design-of-experiment, DOE, based on the single-objective results. The results, together with the initial DOE are shown in Figure 16. This chart includes also the reference Mejzlik 18x6 propeller CR configuration – colored diamonds. The figure depicts initial DOE, thus tradeoffs are visible, these are used to persist the final Pareto frontiers for the 3 design propeller gaps, Z .

After conducting the multi-objective optimization process, the final Pareto frontiers were found and are presented as lines in Figure 16.

The search for the Pareto frontier discovered lower P_e and lower OASPL designs, better than the single-objective cases. This is mainly regarding the minimum OASPL which reaches down to 70.6 dB – some 5 dB below the reference CR system. This was accomplished by getting into very slow operated propeller (about 3,500 rpm) with very high required P_e (more than 2.5 kW) – although these are optimal, it is not a practical design.

The manufacturing and test phases cannot be done for vast amount of propeller designs. Thus, the tradeoff Pareto frontiers can be used to define the optimized configuration to be fabricated. Comparing the electric power and acoustic potential with the current-of-the-shelf Mejzlik 18x6 proves the importance of the optimization process and its advantage. The manufacturing decisions can be taken after the well identification of the optimal tradeoff, with enough supported data. This is accomplished with a well investigated design space which was researched in a thorough and meticulous manner.

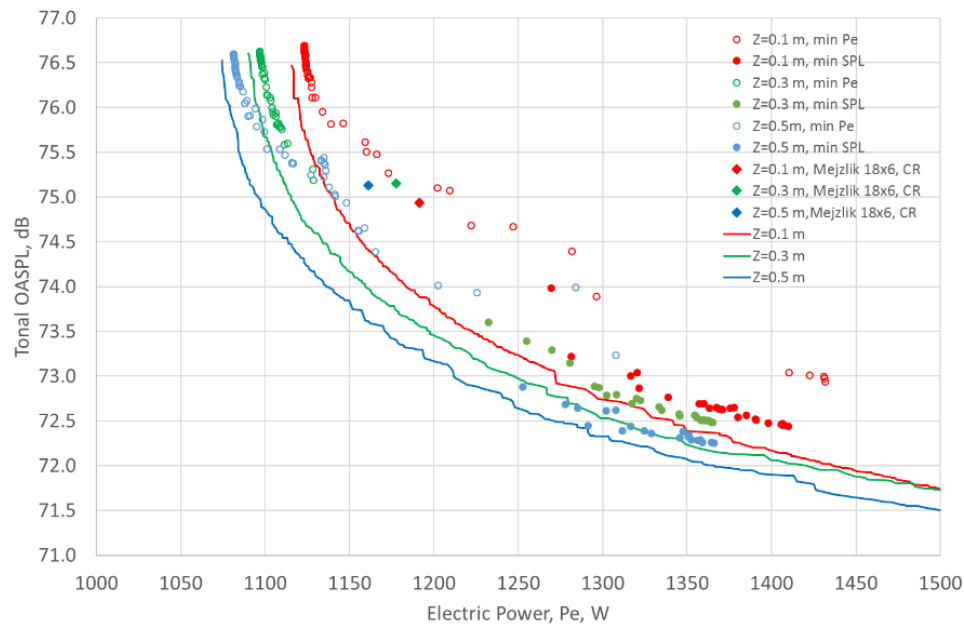


Figure 16, OASPL versus P_e – Final Pareto frontiers

D. Final designs for fabrication

From the above configurations, 3 designs are chosen to be fabricated and tested.

- min. P_e – the best design from the minimum P_e , tradeoff Pareto was taken, as appear in Figure 16.
- min.SPL –the Pareto tradeoff revealed low OASPL designs (down to 70.6 dB). Still these exhibit very high P_e (due to the low rpm). Thus a compromised, practical minimum OASPL was chosen with $P_e=1,350$ W.
- Pareto tradeoff design. A design which shares the same tonal OASPL as the Mejzlik 18x6, but with minimum P_e .

Due to the very small geometric difference between the different Z designs, for all selected designs, the Z=0.5 geometry was chosen. Figure 17 shows the geometries of the 3 propeller systems (total of 6 propellers) with comparison to the reference Mejzlik 18x6. Table 3 shows the numeric values of the various propellers.

As before, the speed is one of the major parameters to improve P_e (increased speed) and OASPL (decreased speed). The increased speed (for better P_e) is accomplished by both decreased pitch and decreases chord. The decreased speed (for minimum OASPL) is accomplished by an increased pitch, increased chord, thus decreased thickness.

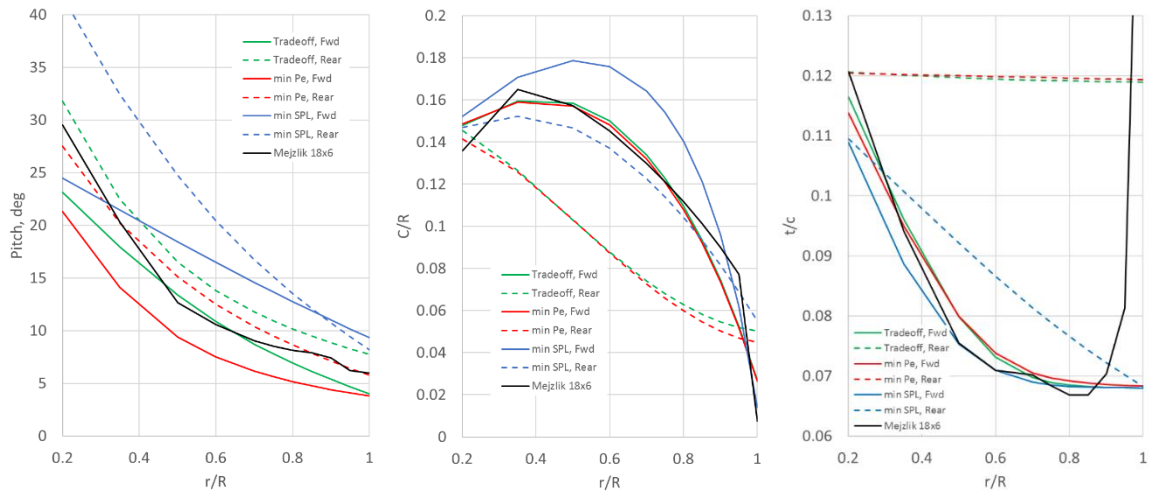


Figure 17, Optimized CR propellers' geometry

Table 3, min P_e from tradeoff Pareto, Comparison of CR propeller systems

Total thrust = 5.6 kgf

	Total P_e , W	OASPL, dB	(Fwd.) Prop. Speed, rpm	$\Delta\Omega$, rpm (Rear-Fwd.)
Mejzlik 18x6	1,160	75.1	5,350	600
min. P_e	1,070	76.5	6,390	20
min.SPL	1,350	72.0	4,500	0
Tradeoff	1,090	75.1	5,830	80

	Thrust, N		P_{e_s} , W		P_{shaft} , W		FM		η_e	
	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear	Fwd.	Rear
Mejzlik 18x6	30.1	24.8	526	635	393	472	0.677	0.421	0.748	0.743
min. P_e	32.8	22.1	564	511	439	404	0.688	0.415	0.779	0.791
min.SPL	30.2	24.7	627	726	415	455	0.644	0.435	0.661	0.627
Tradeoff	33.2	21.7	592	500	445	389	0.693	0.419	0.750	0.778

VIII. Summary and Conclusions

In today urban-air-mobility development, capability of optimized propulsion system design is important. The future electric vertical takeoff and landing vehicle will require propulsion system which are both small and capable of producing high thrust. This makes the contra-rotating propeller configuration as a preferable option to decrease dramatically the size of the vehicle, without decreasing its thrust. Unfortunately, it comes with some performance and acoustic penalties. Through optimization-based design, this penalty can be minimized, which allows high feasibility to contra-rotating configuration.

The paper describes optimization design framework which enables design of contra-rotating propeller system. This framework allows the use of both aerodynamic, acoustic, and electric model, thus enables analyzing full propulsion system capabilities through multidisciplinary design optimization methods.

The various analyses are described in the paper together with relevant validation tests. The validation tests were done using a reference Mejzlik 18x6 propellers and by comparing the analyses results to the experiment, each analysis accuracy was proved.

The aerodynamic analysis is based on expansion of single-rotating blade-element model. The analysis is within 5% accuracy. Acoustic model is based on super-position of the noise signature of the two propellers in the contra-rotating configuration. This attitude does not take into consideration the interaction noise components; thus it might give some deteriorated comparison to tests. Still, for design, the model proves its merit due to the correct trends compared to the experiments. The electric model is based on simple parameterized model which was calibrated versus various tests for the current electric system.

The optimization tools are based on ESTECO modeFRONTIER with capabilities of substantiating the Pareto frontiers, hence finding the full potential and tradeoff of a design problem. The optimization was conducted using hybrid technique with multi-phased optimization scheme. Starting from single objective design problems, which leads to the multi-objective optimizations and finally finds the tradeoff Pareto frontier.

The design results exhibit improvement of electric power consumes by the contra-rotating system and decreased tonal overall sound-pressure-levels. The electric power can be decreased by 7% compared to the reference propeller, and tonal noise can be decreased by 3 dB compared to this reference. Note that the reference propeller is a well-designed propeller, which emphasize the capabilities of the optimization design framework to push design to its full potential.

The results indicate that two single-rotating, side-by-side configuration, are superior compared to an equivalent contra-rotating configuration. Still, by harnessing optimization-based design, this superiority is decreased, thus enabling the usage of smaller configuration, which enables, smaller, simpler, lighter, and cheaper vehicles.

Acknowledgments

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